

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

IV Semester of BSc (BT/MI/BC) Examination April-May 2018

Course code & Name: BS404 Cell Biology- II

Date: 30/04/2018 Day: Monday Time: 10:00 am to 10:20 am Maximum Marks: 10

MCQ

Important Instructions:

- Tick the correct answer and it should be written in question paper itself.
- Use of non-programmable calculator is allowed.

Q - I Choose the correct answer for the following questions. 10

1. _____ is the single most abundant protein in the human body and also major constituent of ECM.
(i) Fibronectin (ii) Collagen
(iii) Laminin (iv) Proteoglycan
2. Which trans-membrane proteins are present in Gap junction?
(i) Connexins (ii) Connectins
(iii) Cadherins (iv) Selectins
3. Number of mitotic division is required to produce 128 cells from a single cell is _____.
(i) 7 (ii) 8
(iii) 32 (iv) 64
4. Which of the following cyclin/CDK regulates progression from G2 to M Phase?
(i) Cyclin D/ CDK4 (ii) Cyclin B/ CDK1
(iii) Cyclin C/ CDK3 (iv) Cyclin E/ CDK2

5. A cancer of _____ is known as Carcinomas.
(i) connective tissues (ii) epithelial cells
(iii) blood-forming cells (iv) all of these
6. Which lysosomal protease gets activated during necrosis?
(i) Calpain (ii) Caspase
(iii) Cathepsin (iv) All of these
7. All cells contain the same genetic information. Why can't cells other than stem cells differentiate into various tissues?
(i) Stem cells are the only cells that can be implanted.
(ii) As cells develop some genes are turned off permanently.
(iii) Stem cells are the only cells that do not have an X or Y chromosome and can therefore go into either a male or female.
(iv) As cells develop their genetic makeup changes.
8. In which signaling pathway receptor dimerization is a key phenomenon for activation of signaling.
(i) Receptor protein-tyrosine kinases (ii) G-Protein coupled receptor
(iii) RAS-MAP kinase (iv) None of these
9. Which of the following chemical agent is stains live cells?
(i) Evan's Blue (ii) Fluorescein Diacetate (FDA)
(iii) Trypan blue (iv) Ethidium bromide
10. Select the correct order of organelle separation based on sedimentation rate during Differential centrifugation.
(i) Nuclei → Mitochondria → Microsome
(ii) Mitochondria → Nuclei → Microsome
(iii) Microsome → Mitochondria → Nuclei
(iv) Nuclei → Microsome → Mitochondria

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

IV Semester of BSc (BT/MI/BC) Examination April-May 2018

Course code & Name: BS404 Cell Biology- II

Date: 30/04/2018 Day: Monday Time: 10:21 am to 12:00 pm Maximum Marks: 25

Instructions:

1. **Section I and II must be attempted in SINGLE ANSWER SHEET.**
2. Make suitable assumptions and draw neat figures wherever required.
3. Use of non-programmable calculator is allowed.
4. Show necessary calculations.

SECTION - I

Q - II Answer the following questions as directed (any five) 10

- (a) What is Extracellular matrix? Enlist the macromolecules involved in organization of ECM.
- (b) Write a note on “Fluorescence-activated cell sorting”.
- (c) Explain in brief about pluripotent and totipotent with examples.
- (d) Give the differences between necrosis and apoptosis.
- (e) Write a note on “Tight Junction”.
- (f) Explain oncogenes and tumor suppressor genes with examples.
- (g) Summarize the event of the first meiotic prophase.

SECTION - II

Q-III Answer the following questions as directed (any five) 15

- (a) Enlist the different second messengers? Discuss the role of cyclic AMP in activation different signaling cascades.
- (b) What carcinogen? Explain the properties of cancer cell.
- (c) Write a brief note on intrinsic pathway of apoptosis.
- (d) Describe in brief about different methods of cell viability and write the application of cell viability testing.
- (e) Explain in detail about arrestins mediated internalization of GPCR”.
- (f) What is Cell–cell interaction? Explain in brief about distinct families of integral membrane proteins involved cell–cell adhesion.
- (g) Discuss the role of RB protein in cell cycle regulation.